



HWA CHONG INSTITUTION
C2 Preliminary Examinations
Higher 3

CANDIDATE
NAME

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CT GROUP

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CENTRE
NUMBER

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INDEX
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PHARMACEUTICAL CHEMISTRY

9812/01

Paper 1

18 September 2015

2 hours 30 minutes

Additional Materials:

Answer Paper
Insert
Data Booklet



READ THESE INSTRUCTIONS FIRST

Write your index number, name and CT on all the work you hand in.

Begin each question on a **fresh** sheet of writing paper. A **nil return** is necessary for any unattempted question.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **five** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

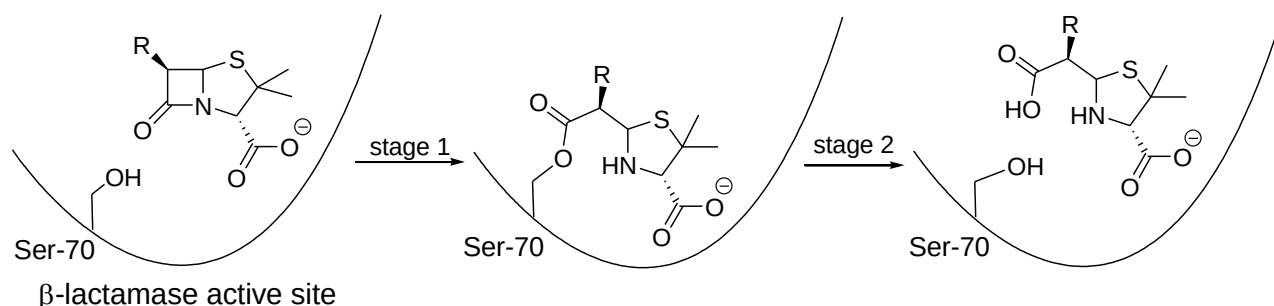
You may use a calculator.

You are reminded of the need for good English and clear presentation in your answers.

1 This question is on antibiotics.

With the prevalent use of penicillin, some strains of bacteria have developed resistance by acquiring the ability to produce β -lactamases. These are enzymes which can hydrolyse and inactivate typical β -lactam antibiotics before they can reach the transpeptidase targets.

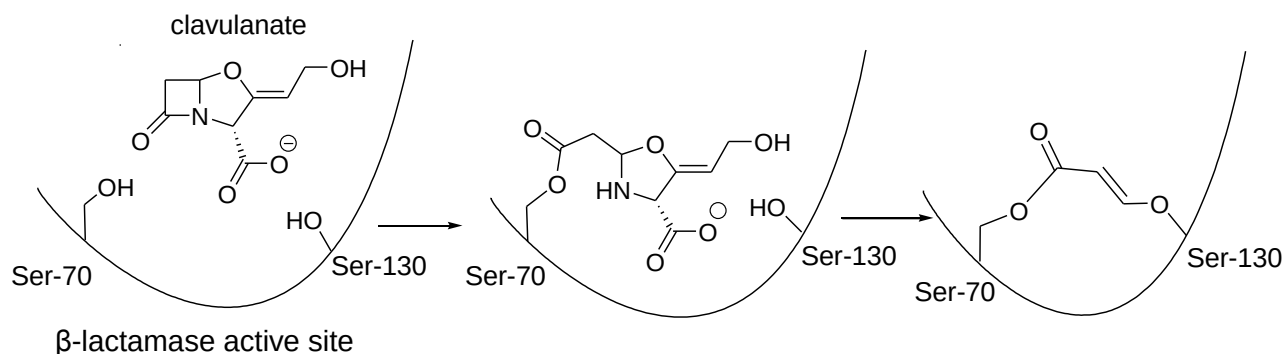
- (a) Explain why the prevalent use of penicillin may have led to resistance. [2]
- (b) Penicillins are inactivated when they form stable adducts with the serine residue in the active site of β -lactamase enzyme. The ester group in the adduct is subsequently hydrolysed to regenerate the free Ser-70 hydroxyl group on the enzyme. [3]



Outline the mechanism of stage 1 and stage 2. [3]

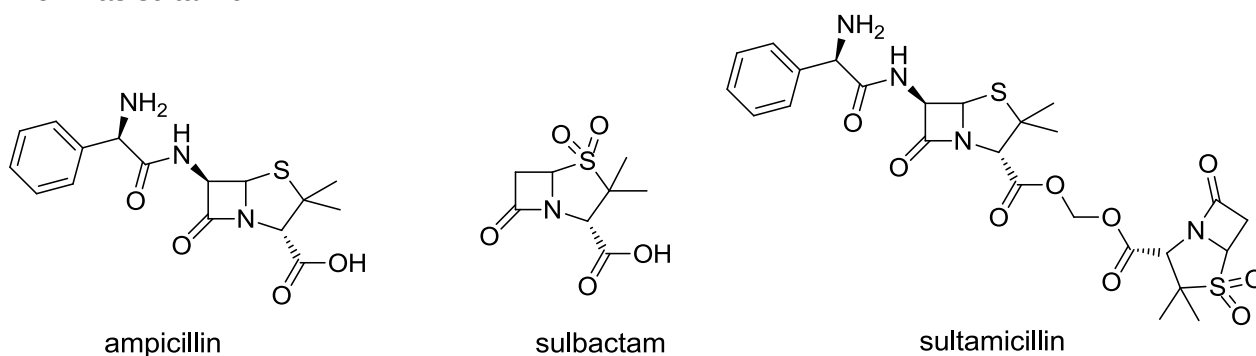
- (c) To overcome antibiotic resistance in bacteria that secrete β -lactamase, penicillin can be given in combination with a β -lactamase inhibitor. Members of this class of competitive irreversible inhibitors include clavulanic acid, sulbactam and tazobactam.

The interaction of clavulanate with β -lactamase is shown below.



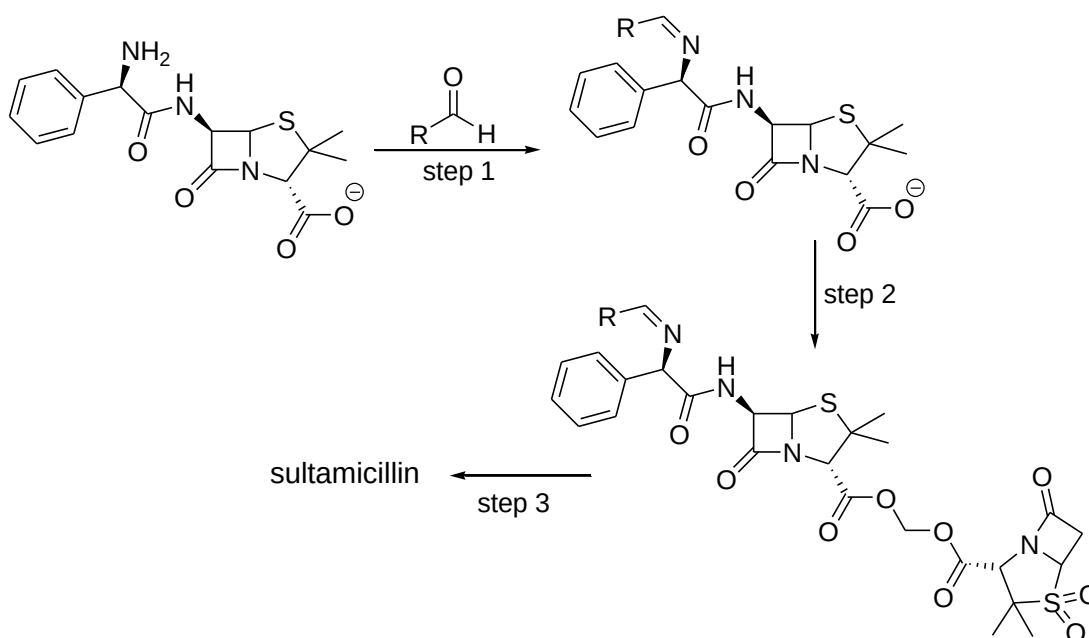
With reference to the structure of clavulanate and its interactions with β -lactamase, explain how it acts as a *competitive irreversible* inhibitor. [2]

- (d) The antibiotic ampicillin can be administered with sulbactam for better activity. They can be administered in two different forms, an intravenous mixture of the two drugs or as an oral prodrug known as sultamicillin.

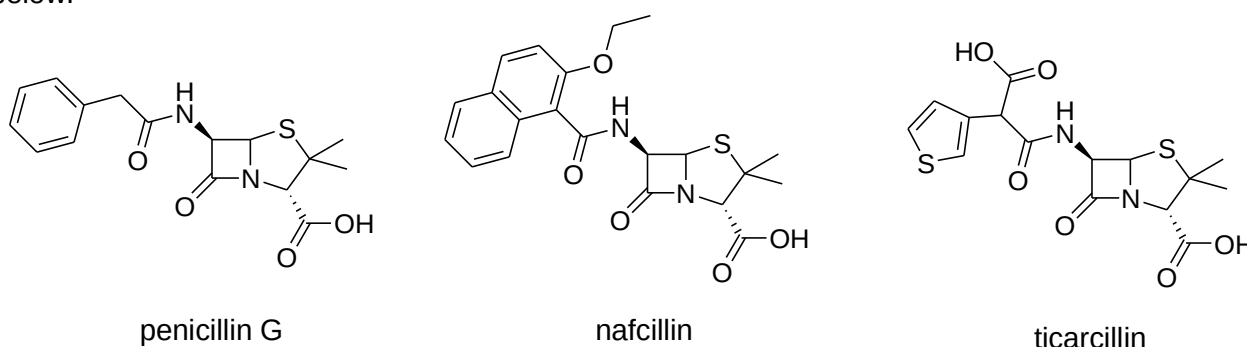


- (i) Suggest reasons for the two different routes of administration. [2]

The synthesis of sultamicillin is as shown below.



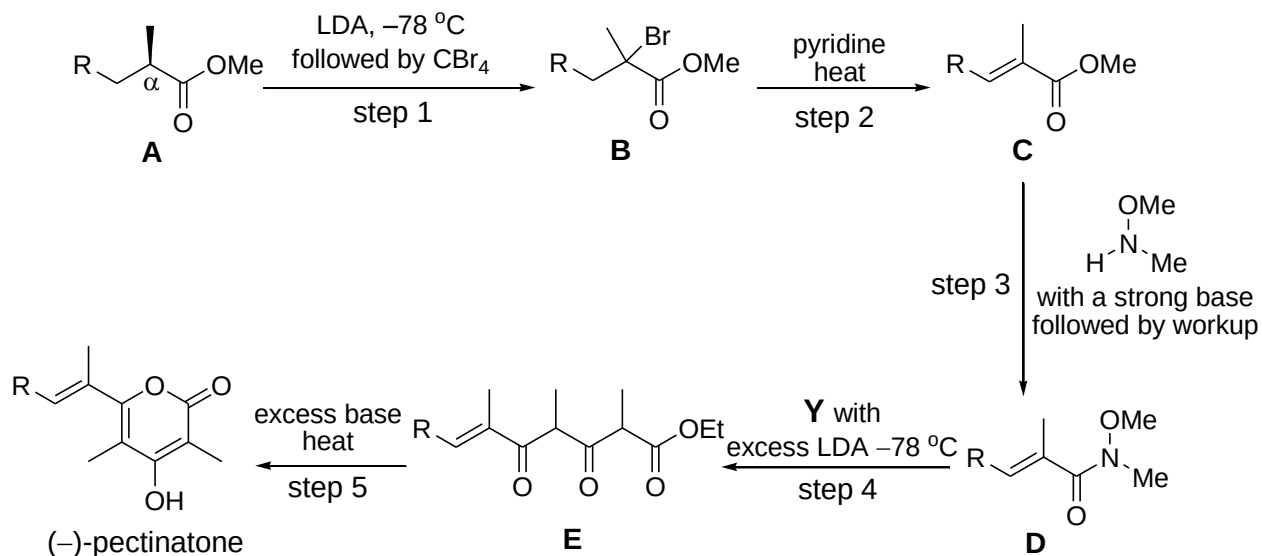
- (ii) Explain why step 1 is necessary. [1]
- (iii) Suggest the reagent for step 2. [1]
- (e) Penicillin G, nafcillin and ticarcillin are other examples of penicillins. Their structures are shown below.



By considering the nature of their respective R groups, rank these three antibiotics in increasing order of susceptibility to intramolecular hydrolysis of the β -lactam ring. Explain your answer. [2]

- (f) Pectinatone was first isolated from the skin extract of the pulmonate *Siphonaria pectinate* collected on the sea wall at the entrance of Key Biscayne. It was found to display antimicrobial activity and cytotoxicity.

The synthesis of (–)-pectinatone is illustrated in the scheme below. For clarity, the strong base lithium diisopropylamide, $(\text{Me}_2\text{CH})_2\text{N}^-\text{Li}^+$, is represented as LDA.

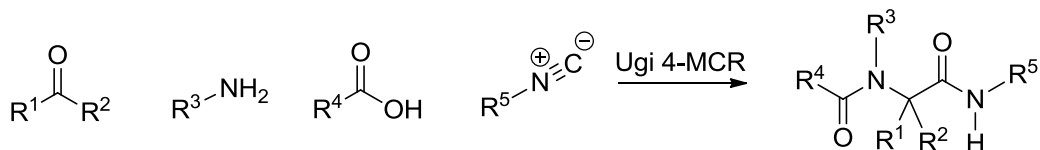


- With the aid of a diagram, explain why the hydrogen on the α carbon is acidic. [1]
- Given that CHBr_3 is formed as a by-product, outline the mechanism of step 1. Hence explain why **B** is obtained as a racemic mixture. [3]
- What *type of reaction* is step 3? Suggest why a strong base is added together with $\text{NH}(\text{OMe})\text{Me}$. [2]
- Draw the structure of reagent **Y** used in step 4. [1]

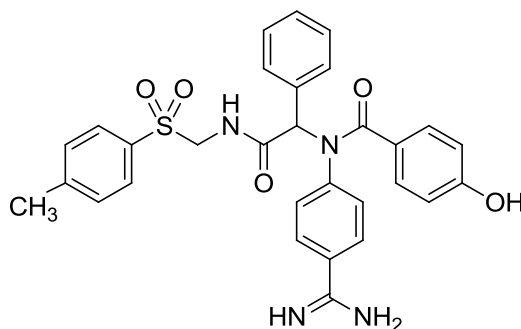
[Total: 20]

TURN OVER FOR QUESTION 2

- 2 Multi-component reactions (MCRs) play a crucial role in combinatorial chemistry for efficient drug discovery. The Ugi reaction is an example of a four-component MCR (4-MCR) involving a carbonyl compound, an amine, a carboxylic acid and an isocyanide to form a bisamide (a compound containing two amide groups).

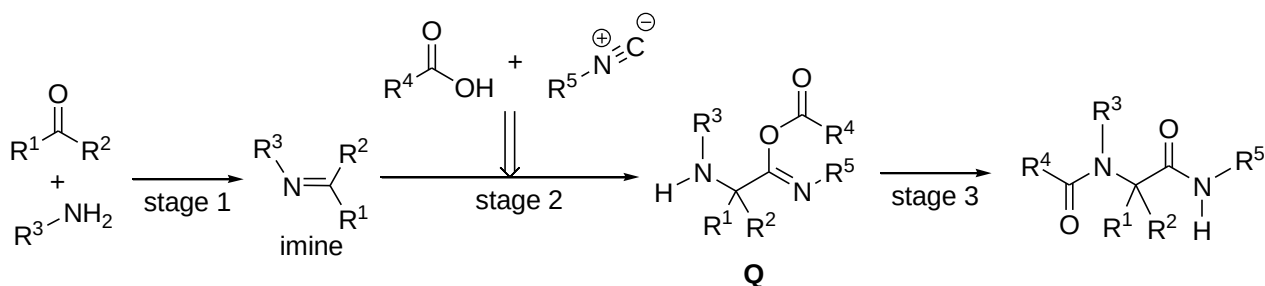


- (a) Draw the structures of the four components that are needed for the synthesis of the thrombin inhibitor, **P** via an Ugi 4-MCR.

**P**

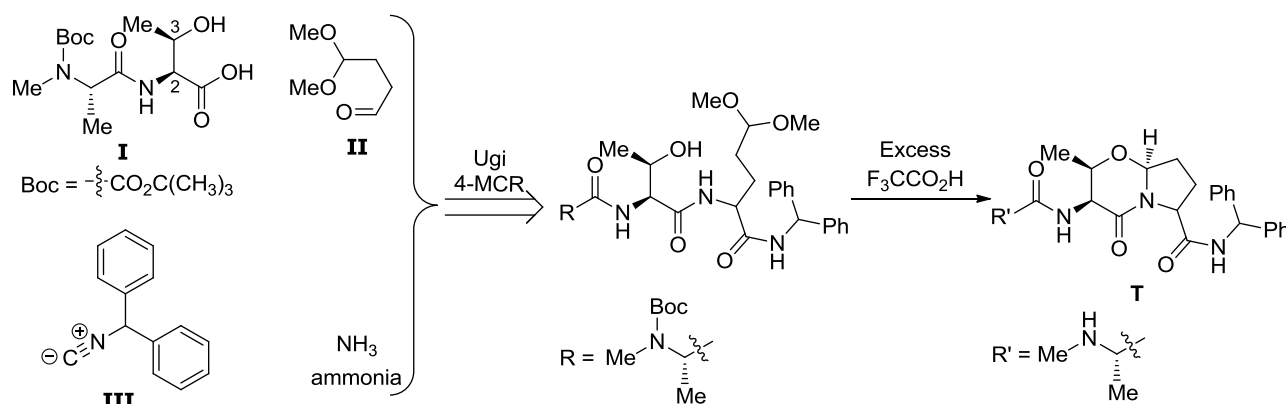
[2]

- (b) The Ugi reaction is thought to first involve the formation of an imine from the carbonyl compound and the amine in stage 1. The imine is then protonated by the carboxylic acid before reacting with the isocyanide to form intermediate **Q**. Rearrangement of the intermediate **Q** gives the bisamide product.



- (i) Draw the mechanism for the formation of intermediate **Q** from the imine in stage 2. [3]
- (ii) Suggest a mechanism for the rearrangement of intermediate **Q** to give the bisamide product in stage 3. [1]

- (c) In 2013, Cosford et al. published a research on how the Ugi reaction may be used as one of the key steps in the synthesis of a class of potent anti-cancer drugs. An initial test involved an Ugi 4-MCR from a Boc protected dipeptide, **I**, a butanedial monoacetal, **II**, an isocyanide, **III**, and ammonia. Subsequent treatment with excess trifluoroacetic acid, $\text{F}_3\text{CCO}_2\text{H}$ produced the lead compound, **T**.



- (i) The product of the Ugi 4-MCR is formed as a 50:50 mixture of a pair of stereoisomers.

Draw the structures of these two stereoisomers and state their relationship. Using your mechanism in (b)(i), explain why two stereoisomers are formed in the reaction. [3]

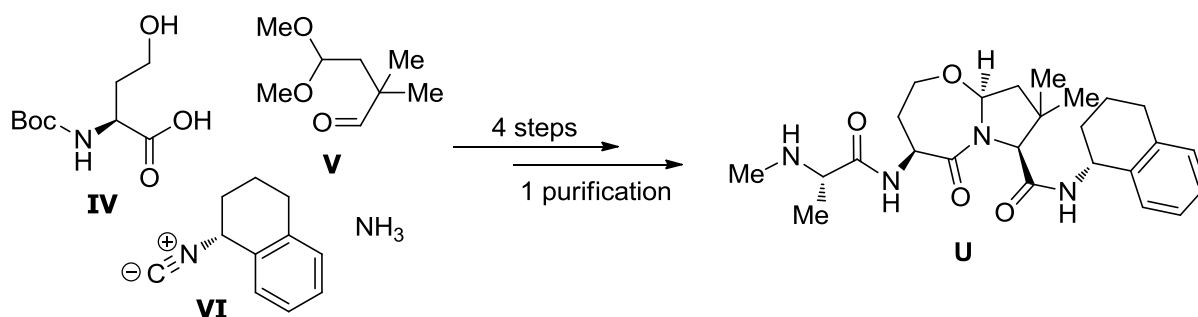
- (ii) Suggest the purpose of protecting the amine group of the N-methylalanine residue in the dipeptide **I**. [1]

- (iii) By considering a 360° rotation along the C2-C3 bond of the dipeptide **I**, draw the Newman projections of the three different staggered and three different eclipsed conformers of **I**. Use R to represent the $-\text{CH}(\text{CH}_3)\text{N}(\text{Boc})\text{CH}_3$ side-chain in **I**. [2]

- (iv) Among the three staggered conformers of **I** in (c)(iii), two conformers are of similar energies and more stable than the third conformer.

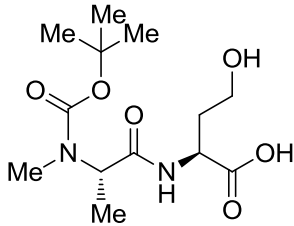
Identify and explain why these two conformers are of similar energies when compared to the third staggered conformer. [1]

- (d) The synthesis of the most potent lead compound, **U**, was achieved by using compounds **IV**, **V** and **VI** for the Ugi reaction.



- (i) Copy the structure of **U** and indicate the R/S configuration of all the chiral carbons. [2]

In the synthesis of compound **X** which is a possible carboxylic acid substrate in the above reaction, an inseparable side product **Y** was formed. It was postulated that **Y** could have been formed from **X** via an intramolecular condensation reaction. The structure of **X**, and the ^1H NMR data for **X** and **Y** are given below.

 X	^1H NMR data of X		^1H NMR data of Y	
	δ	Integration	δ	Integration
	1.38	(d 3H)	1.38	(d 3H)
	1.47	(s 9H)	1.47	(s 9H)
	1.97	(m 2H)	2.32	(m 2H)
	2.86	(s 3H)	2.86	(s 3H)
	3.53	(t 2H)	4.02	(t 1H)
	4.02	(t 1H)	4.30	(t 2H)
	4.41	(q 1H)	4.41	(q 1H)
	5.00	(s 1H)	8.00	(s 1H)
	8.00	(s 1H)		
	11.00	(s 1H)		

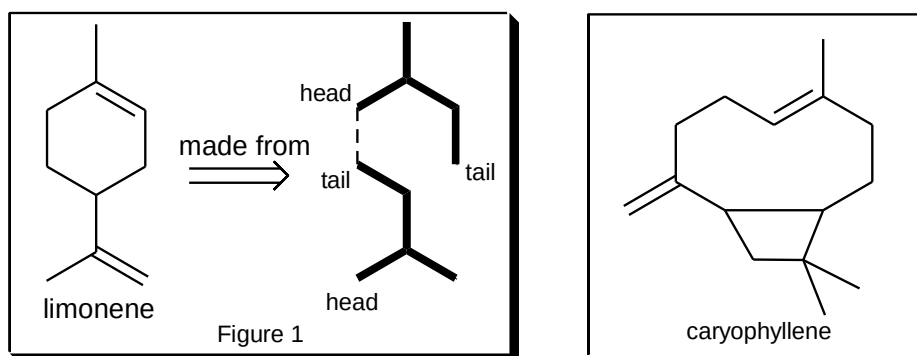
- (ii) Explain how the addition of D_2O may be used to identify the labile protons in **X**. [2]
- (iii) Draw the structure of the side product **Y**. Assign the peaks at δ 2.32, δ 2.86, δ 4.30 and δ 4.41 in **Y**. [3]

[Total: 20]

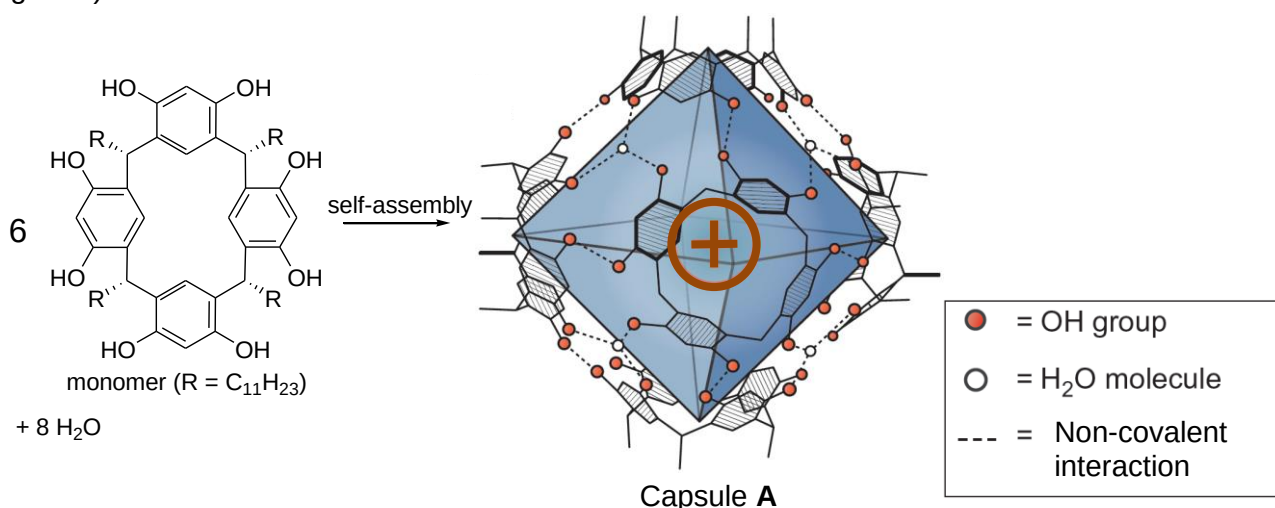
TURN OVER FOR QUESTION 3

- 3 Terpenes are a class of natural products produced primarily by a wide variety of plants and animals. In 1920, the Croatian-Swiss chemist Leopold Ružička developed the isoprene rule which helped to establish the biosynthetic pathway of terpenes.

Using limonene as an example as illustrated in figure 1 below, the 'tail' of one isoprene unit (in bold lines) is joined to the 'head' of another via the dotted line shown. This is how nature produced the terpene backbone; the rest of the σ and π covalent bonds, together with the heteroatoms are added later on by enzymes.



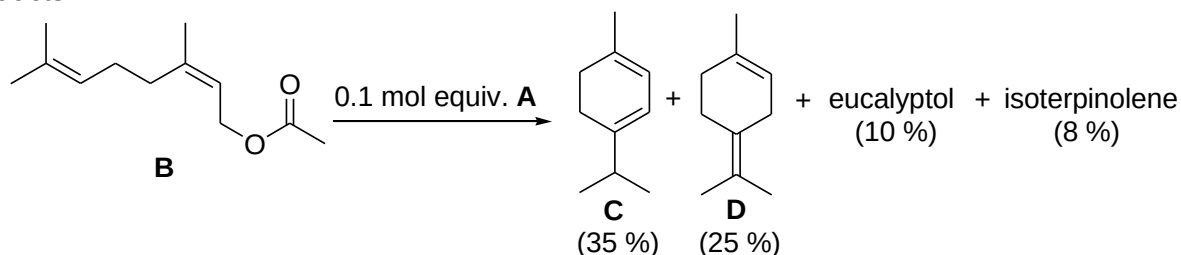
- (a) Caryophyllene is a terpene that is responsible for the citrus scent in black pepper.
- Using the same representation from figure 1, suggest how the caryophyllene backbone is formed in a tail-to-head fashion via **three** isoprene units in its biosynthesis. Label the head and tail end of individual isoprene units. [1]
 - Caryophyllene exists as the (–)-optical isomer and it was also found to be a selective agonist of cannabinoid receptor. Suggest why its enantiomer cannot bind well to the cannabinoid receptor. [1]
- (b) Figure 2 shows a supramolecular capsule **A**, which was obtained from the self-assembly of 6 monomer units and 8 units of water. Capsule **A** is found to stabilise cations (represented with $\oplus$ in figure 2) via cation- π interaction.



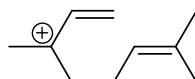
Recently, Tiefenbacher and co-workers studied terpene cyclisation catalysed by capsule **A** and found that it functions as a mild Brønsted acid ($pK_a \sim 6$).

- Suggest the non-covalent interaction that holds the various units together in capsule **A**. [1]
- Hence explain why capsule **A** is a stronger Brønsted acid than the monomer. [1]

- (c) In one experimental study, substrate **B** was mixed with 0.1 mol equivalent of capsule **A** and the reaction was carried out for 75 hours. The scheme below shows the structures of two major products and the names of two of the minor products. The figures in bracket are the yields of the products.



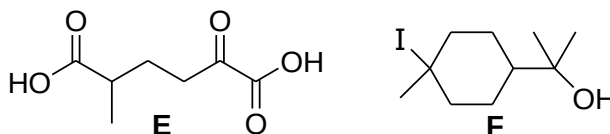
- (i) Illustrate the mechanism for the reaction leading to **C** and **D**, given that the following intermediate was isolated during the course of the reaction. You may treat capsule **A** as a source of protons.



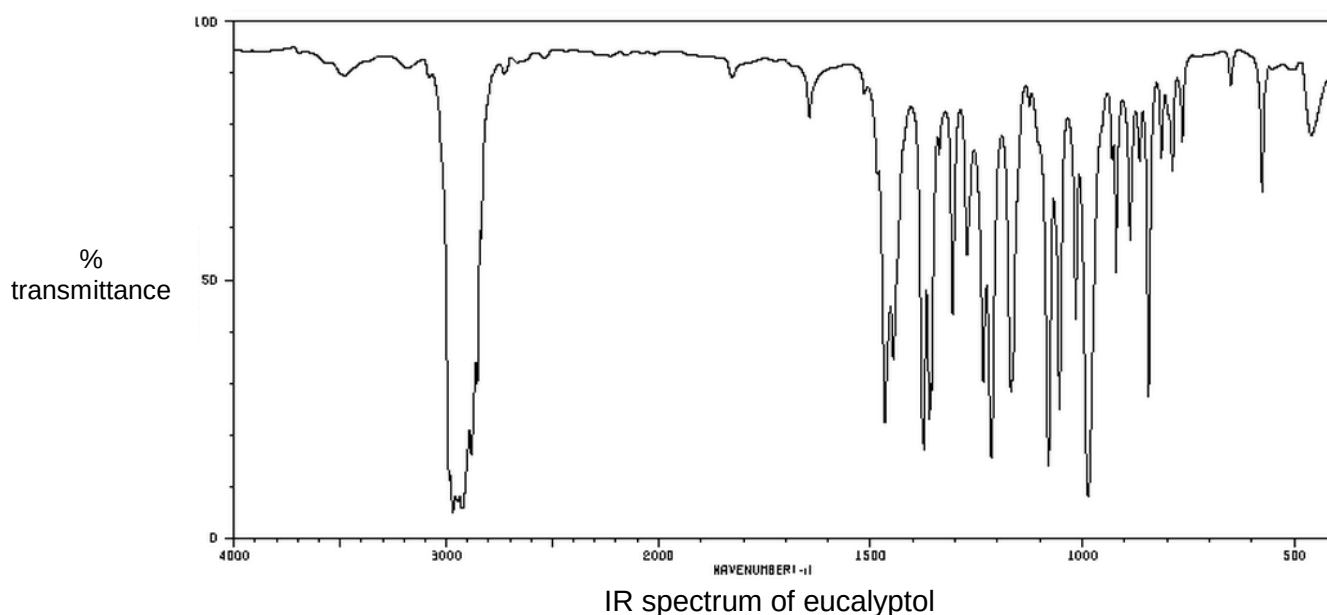
Briefly explain why the yield of **C** is higher than that of **D**.

[3]

Two side products, isoterpinolene ($C_{10}H_{16}$) and eucalyptol ($C_{10}H_{18}O$) were isolated in the earlier experiment. Isoterpinolene was found to have a sharp UV absorbance at around 250 nm whereas there are no significant peaks above 200 nm for eucalyptol. Isoterpinolene is able to rotate plane polarised light whereas eucalyptol is unable to do so. When subjected to hot acidified potassium manganate(VII), isoterpinolene gives CH_3COCH_3 and **E**. When subjected to concentrated HI, eucalyptol gives **F**.

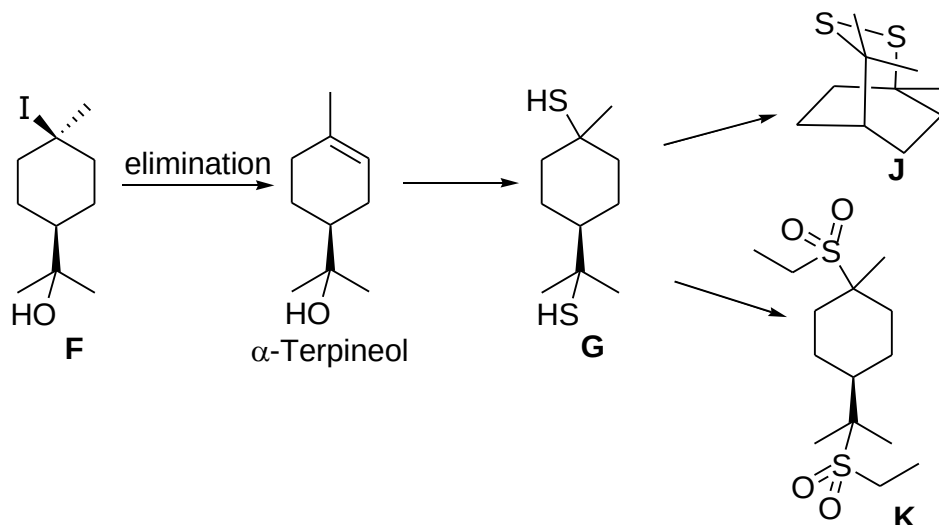


The IR spectrum of eucalyptol is shown below.



- (ii) Outline briefly the **different** principles upon which the techniques of UV and IR spectroscopy are based. Details of instrumentation are not required. [3]
- (iii) Deduce a structure each for isoterpinolene and eucalyptol. Explain how you arrive at your answers. [4]

- (d) α -Terpineol is a derivative of limonene. It can be made from an appropriate stereoisomer of **F** in a one-step elimination reaction. α -Terpineol can be functionalised to **G** which will lead to the eventual synthesis of compounds **J** and **K**. The reaction scheme below summarises the reactions highlighted.

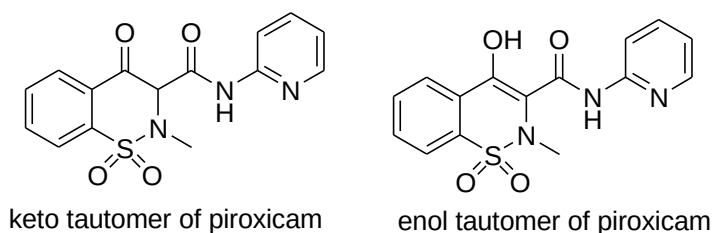


- Suggest an appropriate conformation of **F** that will lead to the production of α -Terpineol via a **one-step** elimination reaction. You should also indicate which hydrogen atom is to be abstracted in your diagram. [1]
- Suggest how **G** could be synthesised from α -Terpineol. Give the reagents and conditions for each step and show each intermediate product. [2]
- Suggest an appropriate stereoisomer of **G** in order for **J** to be synthesised. What reagent is used in this transformation? [1]
- Suggest how **K** could be made from **G**. Give the reagents and conditions for each step and show each intermediate product. [2]

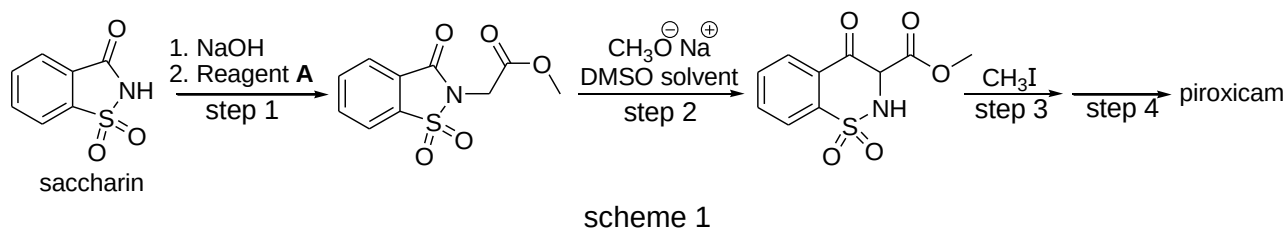
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TURN OVER FOR QUESTION 4

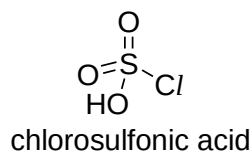
- 4 Piroxicam, $C_{15}H_{13}N_3O_4S$, is a non-narcotic analgesic. It is used to treat painful, inflammatory conditions like arthritis.



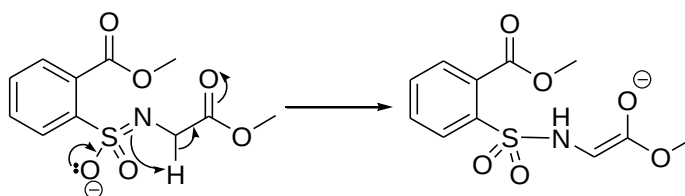
- (a) Outline the mode of action of piroxicam, which is similar to that of aspirin. Contrast the mode of action with that of narcotic analgesics. [2]
- (b) Piroxicam exists in different forms such as keto-enol tautomers and, in polar solvent, as a zwitterion.
- (i) Suggest a reason, other than increased conjugation, why the enol tautomer is more stable than the keto tautomer. [1]
- (ii) Suggest the structure of the zwitterion. [1]
- (c) Piroxicam may be synthesised from saccharin as shown in the following scheme.



- (i) Suggest how saccharin may be synthesised from methylbenzene. Give the reagents and conditions for each step and show each intermediate product. Your synthesis should involve the use of chlorosulfonic acid.

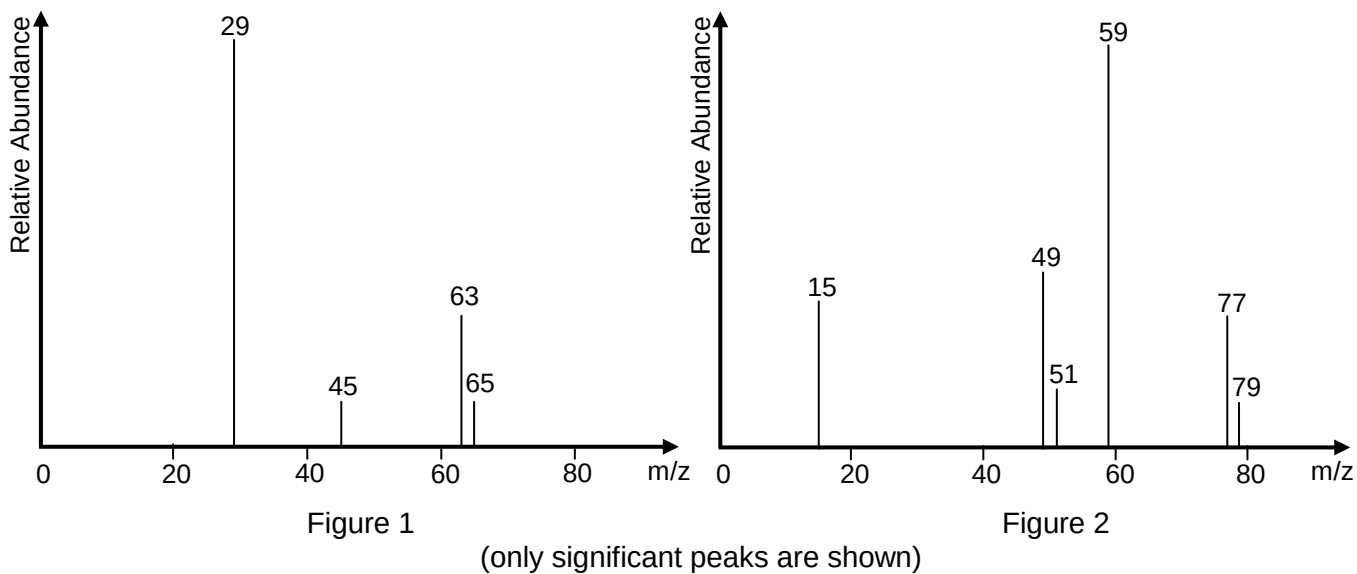


- (ii) Step 2 in scheme 1 is known as the Gabriel-Colman rearrangement. Suggest the mechanism for step 2, given that the following is a key step in the mechanism.



- (iii) Suggest the *type of reaction* in step 3 and the reagent required for step 4. [1]

- (d) Reagent **A** was used in step 1 in scheme 1. Compound **B** is an isomer of **A**. The mass spectra of the two compounds are shown below in Figures 1 and 2, though not necessarily **A** and **B** respectively.

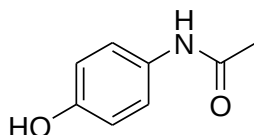


B gives a precipitate on addition of silver nitrate solution.

Suggest structures for **A** and **B**, showing your reasoning.

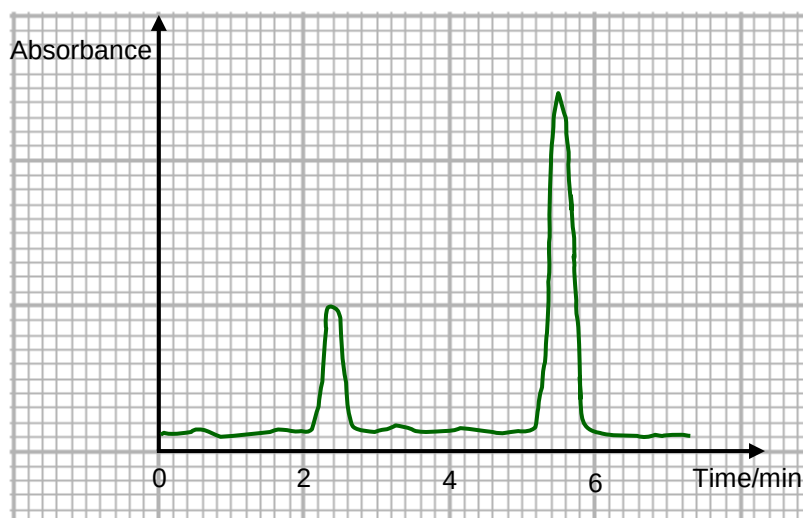
[4]

- (e) Progesic Plus is a tablet containing piroxicam and paracetamol.



paracetamol ($C_8H_9O_2N$)

A sample of Progesic Plus was completely dissolved in a mobile phase of methanol and water. Samples of the resulting solution were analysed by HPLC, and the UV absorption of the eluate was measured at 227 nm. The chromatogram is shown below.



- (i) Suggest which compound was eluted first, using information provided in any part of this entire question to support your answer. [1]

Using standard solutions, under the conditions of the chromatography, peak areas of 1.00 cm^2 for piroxicam and paracetamol were found to correspond to concentrations of 6.0 mg dm^{-3} and 26.2 mg dm^{-3} respectively.

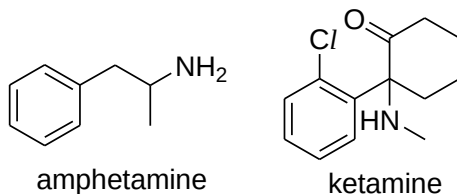
- (ii) Show that piroxicam has a larger molar extinction coefficient than paracetamol at 227 nm. [1]
- (iii) Calculate the relative proportions by mass of each compound in Progesic Plus. [2]
- (f) When piroxicam is reacted with reagent **C**, the prodrug droxicam, $C_{16}H_{11}N_3O_5S$, is obtained. The structure of droxicam is closely related to that of piroxicam.

Given that the ^1H NMR spectrum of droxicam is unchanged upon addition of D_2O , suggest the structure of droxicam and the identity of reagent **C**. [2]

[Total: 20]

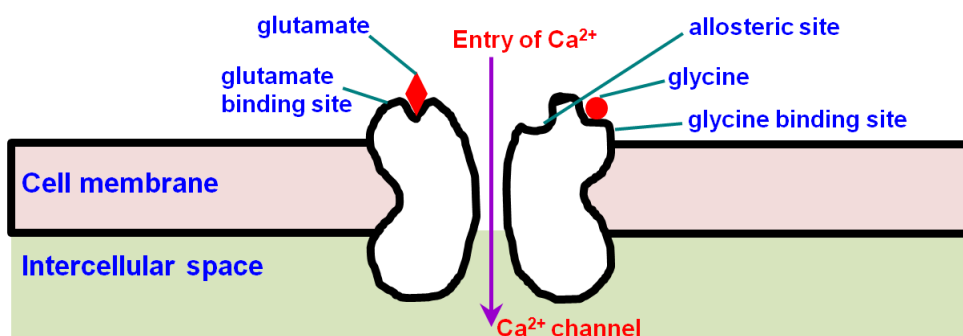
TURN OVER FOR QUESTION 5

- 5 (a) Amphetamine and ketamine are powerful stimulants that are often being misused as recreational drugs.



- (i) Compare and contrast amphetamine and adrenaline, in terms of their chemical structures and their physiological effects. [3]

The N-methyl-D-aspartate (NMDA) receptor is a glutamate-gated cation channel that allows positively charged ions such as Ca^{2+} to flow through when open. A simplified diagram of the activated NMDA receptor, showing the binding of glutamate and glycine to the relevant binding sites is shown below.



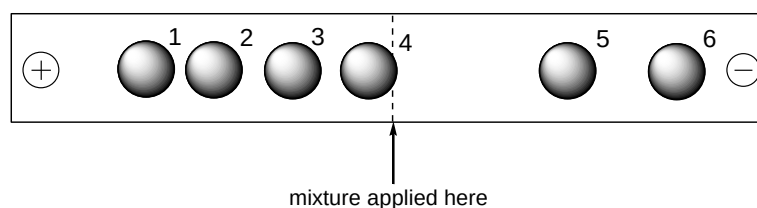
- (ii) Ketamine is an example of a non-competitive NMDA receptor antagonist. Suggest, with the aid of a diagram, how the NMDA receptor will look like when ketamine binds to the NMDA receptor. [2]
- (b) Bacitracin is a cyclic peptide produced by some microorganisms. Partial hydrolysis of bacitracin produced several fragments, six of which are the following. The figures in brackets are the isoelectric point of the compounds.

tripeptide: His-Phe-Ile (7.6);
 dipeptide: Ile-Lys (9.7); Phe-Ile (5.8);
 single amino acids: Ile (5.8); Phe (5.8); Lys (9.7).

The following table gives the structures of the four amino acids involved.

name	Phenylalanine	Isoleucine	Histidine	Lysine
abbreviations	Phe	Ile	His	Lys
structure of side chain				

When a mixture of the above six compounds was subjected to electrophoresis in a buffer at pH 8, the following results were obtained.



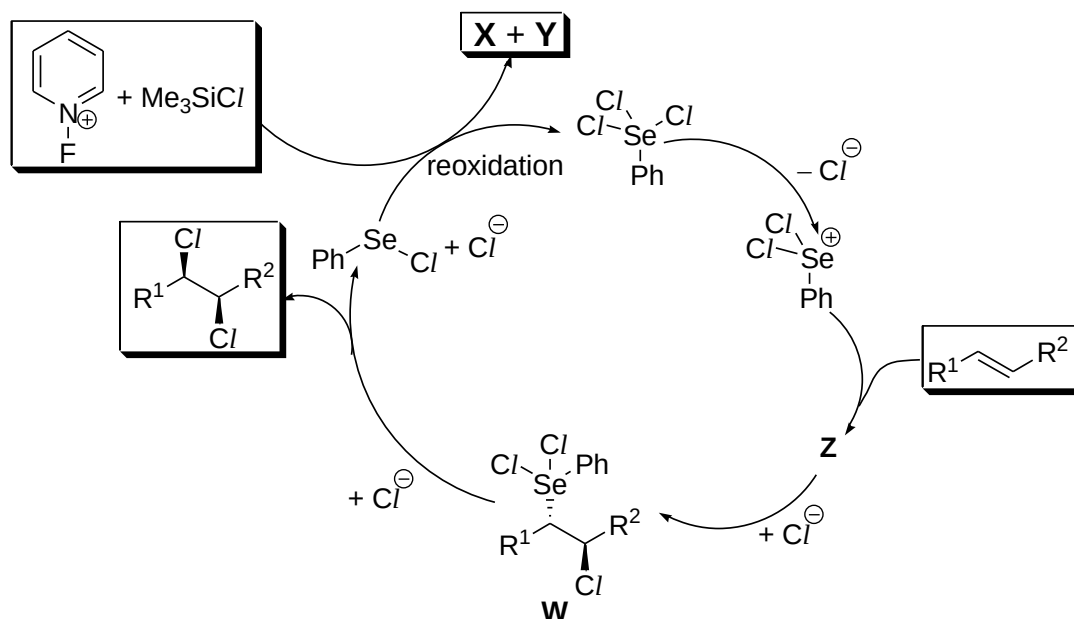
Identify which of the six compounds is responsible for each of the six spots 1–6. [3]

(c) The following part of this question deals with electrophilic addition.

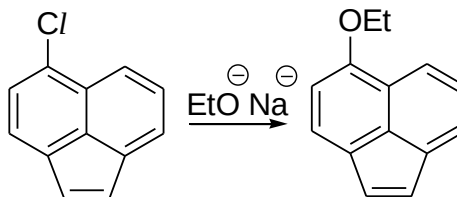
- (i) Electrophilic addition of bromine is often described as an *anti-addition* and the reaction was known to be *stereospecific*.

Using both *cis*-pent-2-ene and *trans*-pent-2-ene together with appropriate curly arrows to depict the movement of electron pairs, explain the two italicised terms above. Appropriate annotations may be useful for your answers to be complete. [3]

Denmark and co-workers recently uncovered a catalytic *syn-addition* of chlorine for a variety of alkene substrates using a selenium based catalyst, PhSeCl_3 . The proposed catalytic cycle is shown below.



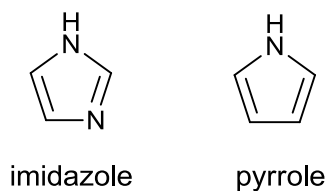
- (ii) Suggest the structure of intermediate **Z** and propose via a suitable mechanism, how **W** could lead to the production of the *syn*-addition product. [3]
- (iii) In the reoxidation step, Se is being oxidised from +2 to +4. Identify the element that is being reduced in the step, stating clearly its initial and final oxidation state. [1]
- (iv) Hence or otherwise, propose the structures of the organic by-products, **X** and **Y**. [2]
- (d) (i) When subjected to sodium ethoxide, chlorobenzene is unreactive whereas 2-chloropyridine reacts readily. Briefly explain why. [2]
- (ii) Suggest with the aid of a diagram why the following reaction occurs much more readily than the reaction between ethoxide and chlorobenzene.



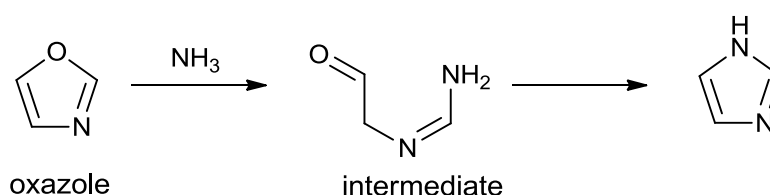
[1]

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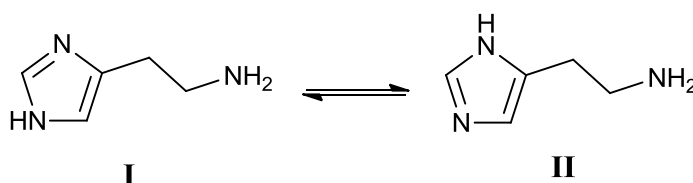
- 6 (a) Many pharmaceutical molecules contain the imidazole ring system.



- (i) Suggest the structures of the conjugate acid derived from the protonation of imidazole and pyrrole respectively. By considering the stability of the conjugate acids, suggest how the basicity of imidazole compares to that of pyrrole. [2]
- (ii) Explain how the acidity of imidazole compares to that of pyrrole. [1]
- (b) Oxazole is a heterocyclic compound that resembles imidazole. It is possible to convert oxazole to imidazole by reaction with ammonia.

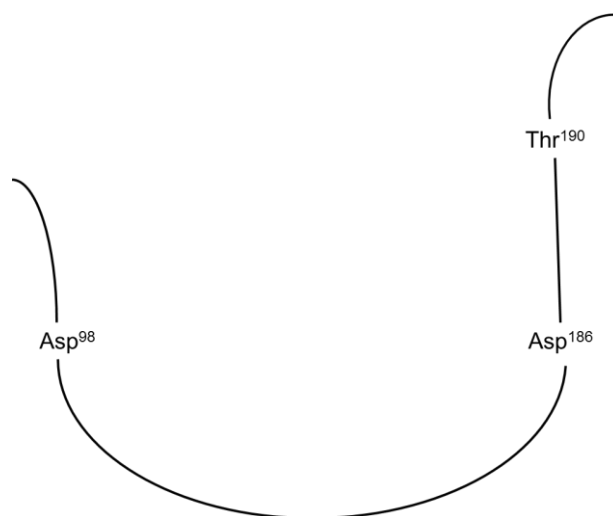


- (i) Suggest a mechanism for the conversion of oxazole to imidazole. [3]
- (ii) Predict how the reactivity of oxazole towards ammonia will change by alkylation at the nitrogen atom (also known as *N*-alkylation). Explain your answer. [1]
- (iii) While imidazoles can be easily *N*-alkylated using methyl iodide, oxazoles require much stronger alkylating agents. Suggest a reason why. [1]
- (c) Histamine is an organic molecule found in the human body that is responsible for a host of biological actions such as local immune responses and regulation of gastric acid secretion through interactions with one of three receptors (H_1 , H_2 and H_3). Histamine exists in two tautomeric forms as shown below. Tautomeric form **II** has been found to be the active form when bound to the H_2 receptor.



- (i) Early pharmacological studies demonstrated that the biologically active form of histamine was a singly protonated monocation (pK_{b1} of histamine = 4.2 at 37 °C). By considering the basicity of the nitrogen atoms in histamine, draw the structure of the monocation of tautomer **II**. [1]
- (ii) Calculate the ratio of the concentrations of singly protonated tautomer **II** to neutral tautomer **II** in the human body at physiological pH of 7.4. You may assume that the temperature of the human body is 37 °C and that the pK_w of water is 13.6 at 37 °C. [2]

- (iii) It is postulated that histamine binds to the H_2 receptor through a three-site interaction. The diagram below shows the binding site of the H_2 receptor. The structures of some amino acids are also displayed.



amino acid	structure
aspartic acid (Asp)	<chem>OC(=O)CC(N)C(=O)O</chem>
threonine (Thr)	<chem>CC(O)C(N)C(=O)O</chem>

By considering the structures of the amino acid residues at physiological pH, draw a diagram to show how histamine binds to the binding site of the H_2 receptor, labelling all interactions clearly. [2]

- (d) Drugs known as antihistamines that act on the H_2 receptor are also known as inverse agonists. Unlike antagonists, inverse agonists bind to the same binding site as the agonist, but produce a pharmacological response **that is opposite** to that of an *agonist*.

- (i) Explain what is meant by the term *agonist*. [1]
- (ii) Sketch on a same graph, how pharmacological response changes with increasing concentrations of agonist, antagonist and inverse agonist respectively. You may assume that the number of H_2 receptors is limiting. [2]

- (e) For this part of the question, you may assume that histamine exists as tautomer **I**.

Once histamine is formed, it is either stored or metabolised by enzymes to form metabolites **A**, $C_6H_{11}N_3$, and **B**, $C_5H_6N_2O$. The action of the enzyme HMT on histamine involves a modification on the imidazole ring to form **A**, while the enzyme DAO modifies the side chain of histamine to form **B**.

Three sets of proton NMR signals are shown below (signals of labile protons are not shown).

Data set X		
δ_H	Integral	splitting
3.02	2	t
3.29	2	t
7.14	1	s
7.99	1	s

Data set Y		
δ_H	Integral	splitting
2.92	2	t
3.25	2	t
3.69	3	s
7.00	1	s
7.70	1	s

Data set Z		
δ_H	Integral	splitting
3.66	2	d
7.14	1	s
7.99	1	s
9.72	1	t

- (i) Suggest which set of proton NMR data corresponds to histamine, and assign the chemical shifts to the protons on histamine. [2]
- (ii) Suggest the structures of metabolite **A** and metabolite **B**. [2]

[Total: 20]

Copyright Acknowledgements:

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- Question 5c © A. J. Cresswell, S. T-C. Eey & S. E. Denmark. Catalytic, stereospecific *syn*-dichlorination of alkenes. *Nature Chemistry*. **7**, 146–152 (2015).